

Server / Power Rack Cold Plate Engineering Development

建立可量產、可驗證、可模組化的 Rack-level 散熱工程平台

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可量產

Design for Manufacturing
從 Concept 到 NPI 標準化流程

可驗證

Thermal / Fluid / Mechanical /
Reliability 四維測試矩陣

可模組化

Chip → Rack → CDU 層層解耦
支援多元應用場景

系統層級散熱需求與 Cold Plate 設計目標

Cold Plate 設計必須同時滿足熱性能、流體力學、機械可靠、與可維修性四維度目標

Category	Target Value	Test Method	Status
Heat Load	300–800W per GPU chip (TBD)	Calorimetry	In Progress
Flow Rate	5–20 LPM per cold plate	Flow meter	Confirmed
Pressure Drop	< 30 kPa @ 15 LPM (TBD)	Differential pressure gauge	In Progress
Thermal Resistance	< 0.05°C-in²/W (TBD)	JEDEC thermal test	Not defined
Leak Rate	Zero leak — 100% helium test	Helium leak detector	Confirmed
MTBF	> 40,000h (CDU side)	Field / accelerated life test	TBD
Operating Temp	-40°C to +85°C (TBD)	Thermal cycling	TBD
Serviceability	Repair time < 30 min (target)	Time study	TBD

(TBD) = To Be Defined — 需要與 RD / PM / CSP 確認後更新 | Status colors: Green=Confirmed / Orange=In Progress / Red=Not defined

從 Chip-level 到 Rack-level 的四層熱源架構

不同熱源有不同散熱介面與設計約束，需針對每層建立對應的 Cold Plate 解決方案

L1 Chip-level — GPU / ASIC

熱通量 $> 500\text{W}/\text{cm}^2$ ，局部熱點明顯。介面：TIM $\rightarrow$ Cold plate base， $R_a < 0.8\mu\text{m}$ ，接觸壓力均勻分布。

L2 Power Module — IGBT / SiC

熱通量 $200\text{--}400\text{W}/\text{cm}^2$ ，面積 $50\times 50\text{mm}\text{--}100\times 100\text{mm}$ 。介面：TIM 或焊接，雙面散熱可選。需電氣絕緣 $> 3\text{kV}/\text{mm}$ 。

L3 Power Rack — 48V Busbar / 400V Dist.

熱密度中等，散熱區域均勻。介面：整合至 busbar 結構，鋁合金擠型散熱片。與電源系統整合優先。

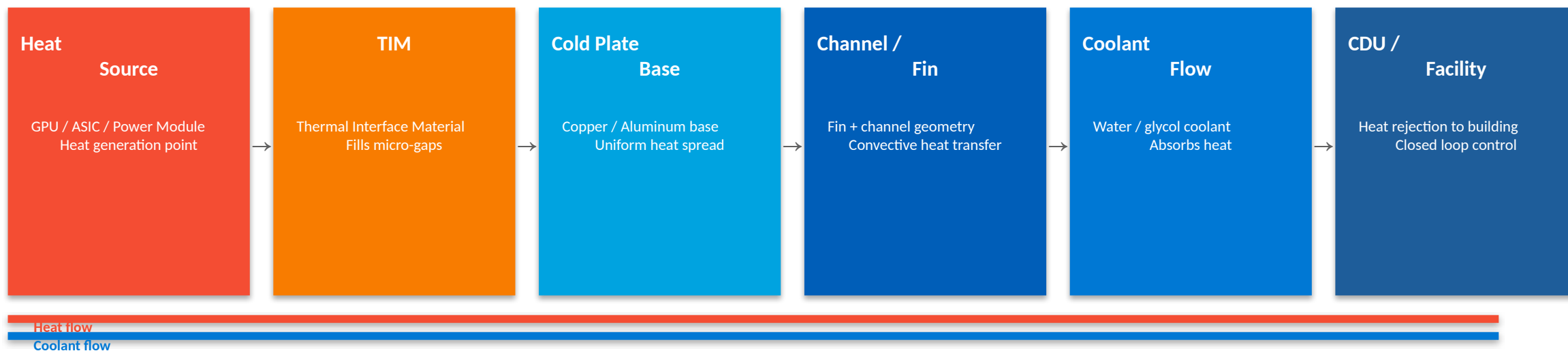
L4 Rack-level — Manifold / Cooling Plate

分配冷卻介質至多個 Cold Plate。介面：Quick disconnect / Manifold。流量平衡、壓力損失一致性設計重點。

每層熱源有不同的設計約束與驗證方法，需建立對應的 Cold Plate 產品線，而非以單一設計覆蓋全部

熱傳路徑從晶片到設施的完整架構

Cold Plate 是熱從晶片到設施冷卻回路的關鍵節點，每一層都有其工程意義



Key Design Parameters

- Fin height / pitch / width
- TIM selection & thickness
- Hydraulic diameter
- Inlet / outlet $\Delta T < 10^{\circ}\text{C}$
- Surface roughness R_a
- Contact pressure distribution

五種主流流道設計方案的技術比較

不同流道幾何適用於不同熱密度與流量條件，需依應用場景選擇

Design	Advantage	Heat Flux Range	Complexity	Recommendation
Parallel Channel	Low δP , uniform flow, easy to manufacture	Medium heat flux < 300W/cm ²	Low	Recommended (Power rack 48V)
Serpentine	Long path, good for high flux, slight δP penalty	Medium-high 300~450W/cm ²	Medium	Applicable (GPU cold plate)
Pin Fin / Skived Fin	High area density, efficient heat transfer	High heat flux > 400W/cm ²	High	Applicable (GPU cold plate)
Microchannel	Ultra-high HTC, minimal thermal resistance	Very high flux (limited by clogs)	Very High	Research only (Mass production TBD)
Hybrid Channel	Combines geometries, optimized for AI server	High density 200~500W/cm ²	Medium-High	Preferred (Primary target)

Geometric diagrams (representative only — detailed CAD geometries to be defined in Phase 1)

Cold Plate 設計四類關鍵參數

設計驗證需從 Thermal / Fluid / Mechanical / Reliability 四個維度同步展開

Thermal	Fluid	Mechanical	Reliability
R_th (熱阻) < 0.05°C-in²/W TBD	Flow rate 5–20 LPM per cold plate Confirmed	Flatness < 0.05mm Confirmed	Thermal cycling 5,000 cycles -40°C/+85°C TBD
Chip ΔT < 5°C across chip TBD	Pressure drop < 30 kPa @ 15 LPM In Progress	Thickness < 6mm for GPU cold plate TBD	Pressure test 1.5× max operating pressure TBD
ΔT (inlet/outlet) < 10°C at design flow Confirmed	Flow uniformity < 10% deviation per channel TBD	Material strength Al 6061-T6 yield > 270 MPa Confirmed	Vibration Per rack environment (TBD) TBD
Contact pressure 0.5–2 MPa uniform TBD	Reynolds number Re < 2300 (laminar) or Re > 4000 (turb) TBD	Mount interface Compatible with rack holder TBD	Corrosion Per water quality spec (TBD) TBD

材料與製程評估 — 散熱性能 vs. 成本 vs. 可量產性

材料與製程的選擇決定了 Cold Plate 的散熱性能上限與量產成本，需在設計階段早期確認

Material	Thermal Conductivity	Cost	Mass Production	Application
Copper (Cu)	~400 W/m·K	High	CNC / Brazing	High-power GPU cold plate
Aluminum (6061/6063)	~150~180 W/m·K	Medium	FSW / Stamping / CNC	Medium-power / Power rack
Stainless Steel (SS)	~15 W/m·K	Medium	Stamping / Welding	Corrosion-sensitive applications
Hybrid (Cu base + Al fin)	~250~300 W/m·K effective	Medium-High	Brazing / FSW	Primary choice — high power AI server

Manufacturing Process Comparison

CNC Machining Prototype / low volume	Brazing (釐焊) High-power cold plate (primary)	FSW (攪拌摩擦焊) High-volume AI production
Skiving Pin fin cold plate (high performance)	Stamping Power rack cold plate (simple geometry)	Surface Treatment Corrosion protection + appearance

密封設計與漏液風險管理

漏液是液冷系統的頭號風險，密封設計必須從架構層面而非零件層面來處理

Seal Type	Cost	Leak Risk	Temp Range	Serviceability
O-ring	Low	Low (periodic replacement)	-40°C to +120°C	Easy — replaceable
Gasket	Medium	Medium (bolt torque critical)	-40°C to +150°C	Moderate
Welding / Brazing	High	Near-zero (permanent)	Wide range	Not serviceable
Compression Fitting	Medium	Low (if assembled correctly)		

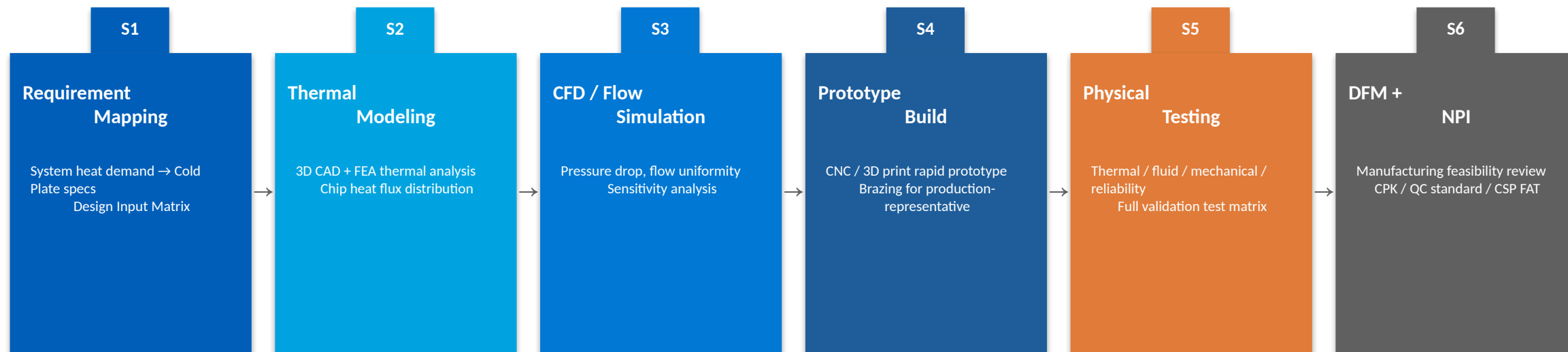
Leak Risk Locations & Mitigation

- Cold plate body seam: Braze/weld → 100% He test
- Quick disconnect interface: O-ring + bubble sensor
- Manifold joints: Gasket + leak detection
- CDU connection: Compression fitting + pressure monitor

Serviceability target: Repair time < 30 min | Quick disconnect with hot-swap capability required

從 Requirement 到 DFM 的標準工程驗證流程

Cold Plate 開發需要一個標準化的驗證流程，確保 design decision 有據可查、有數字支撐



Iteration loop: Failed tests return to Step 2 (Thermal/Fluid redesign) — target max 3 iterations per design cycle

Decision Gates: S1→S2 (Design Review) | S3→S4 (Prototype Go/No-Go) | S5→S6 (DFM Sign-off) | S6→Production (NPI Release)

Cold Plate 驗證測試矩陣 — Thermal / Fluid / Mechanical / System

四類測試構成完整的驗證框架，缺任何一類都無法確保產品達到量產品質

Thermal	Fluid	Mechanical / Reliability	System-Level
R_th vs. flow rate → Pass: within 15% of simulation	P-Q curve (pressure vs. flow) → Pass: < 30kPa @ 15LPM	Pressure hold test → Pass: no leak at 30 min	CDU integration test → Pass: stable feedback control
Surface ΔT (IR imaging) → Pass: ΔT < 5°C	Flow uniformity (array) → Pass: deviation < 10%	Burst / hydrostatic test → Pass: > 3× rated pressure	QD hot-plug test → Pass: 100 cycles, no leak
Inlet/outlet ΔT → Pass: ΔT < 10°C @ design flow	Re number assessment → Confirm design assumption	Thermal cycling → Pass: no cracks, leak rate < 1×10⁻⁸ Pa·m³/s	Rack-level leak test → Pass: zero leak under working pressure
72h steady-state test → Pass: stable operation, no degradation	Water quality corrosion test → Pass: no corrosion after 1,000h	Vibration test → Pass: no mechanical degradation	CSP FAT (customer) → Per customer spec (TBD)

六大技術風險與對應策略

風險需在設計階段早期識別，否則進入 NPI 後的補救成本將呈指數增加

R1 Pressure Drop Exceeds Target

→ CFD override → channel geometry optimization → set δP limit in design rule

Consequence: CDU pump overload, flow starvation, overheating

L:H

I:H

R2 Thermal Target Unmet

→ Build simulation → prototype → test fast iteration capability; TAT < 4 weeks

Consequence: Chip throttling, customer complaint, program delay

L:M

I:H

R3 Leakage Risk

→ Seal design review, 100% He test in production, leak sensor standard

Consequence: Data center incident, liability, brand damage

L:L

I:H

R4 Corrosion / Water Quality

→ Establish water quality spec; material selection (Ni-plated Cu) upfront

Consequence: Channel clogging, efficiency decline, high warranty cost

L:M

I:M

R5 Manufacturing Yield Risk

→ DFM review early in design; early supplier engagement

Consequence: Higher unit cost, NPI delay, supply shortage

L:M

I:M

R6 Field Service Difficulty

→ Modular design, QD standard, service manual, spare parts policy

Consequence: Longer downtime, high post-sale cost, low satisfaction

L:L

I:M

L = Likelihood | I = Impact | H=High / M=Medium / L=Low | RAG: Red=High risk / Orange=Medium

12 個月開發計畫 Roadmap — 從 Concept 到 NPI

三階段里程碑，確保每一階段都有可驗證的交付物，逐步建立 Cold Plate Engineering Platform



Resources required: Thermal/Fluid simulation tools (CAE), CNC prototype capability, Test equipment (TBD in Phase 1)